ZEPTO TRAFFIC INSIGHTS ANALYTICS

Project Report File

WEB ANALYTICS LAB

24CAP-752

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1. Introduction

The **Zepto Web Analytics Dashboard** is a comprehensive, single-page interactive web application developed as the final project for the Web Analytics Lab course (24CAP-752) at the University Institute of Computing, Chandigarh University. The project is built entirely using core front-end technologies — HTML5, CSS3, and JavaScript — with Chart.js for data visualization.

The dashboard presents an in-depth analytics study of **Zepto.com**, one of India's leading 10-minute grocery delivery platforms. It covers five key analytical domains: web traffic analysis, SEO performance, advertising campaign metrics, return on investment simulation, and data-driven growth recommendations.

A standout feature of the project is its fully functional **Dark and Light mode toggle**, implemented using CSS custom properties (variables). The entire visual theme of the dashboard switches instantly with a single click, with all Chart.js charts rebuilding themselves with theme-appropriate colors. The application is fully responsive and requires no server, build system, or installation — it runs directly in any modern browser.

2. Objective

The primary objective of this project is to apply web analytics concepts learned during the lab course by building a real, functional analytics dashboard that demonstrates mastery of front-end development, data visualization, and digital marketing analysis.

Specific Objectives

- To design and develop a professional multi-section analytics dashboard using HTML5, CSS3, and JavaScript without any external frameworks.
- To implement a fully functional Dark/Light mode theme system using CSS custom properties that instantly updates all UI elements and charts.
- To present real web analytics data for Zepto.com including traffic metrics, SEO scores, backlink analysis, and keyword rankings.
- To create an interactive ROI Calculator with live-updating sliders that compute key advertising metrics in real time.
- To visualize campaign performance data using Chart.js with dynamic, theme-aware charts for Traffic Trends, Campaign CTR vs CVR, and ROI curves.
- To demonstrate practical knowledge of advertising KPIs — CTR, CPC, CAC, ROAS, ROI — with actual formula calculations.

3. Technologies Used

Technology	Version / Source	Role in Project
HTML5	Standard	Page structure, semantic markup, all dashboard content
CSS3	Standard	Custom properties for theming, layout grids, animations
JavaScript	ES6+	Tab navigation, ROI calculator logic, chart rebuilding
Chart.js	v4.4.1 via CDN	Three interactive charts: line, bar, and ROI curve
Google Fonts	Outfit + JetBrains Mono	Display typography and monospace number formatting
CSS Variables	Custom properties	Complete dark/light mode — 14 token pairs per theme

4. Project Structure and Features

The entire project is contained in a **single HTML file** (zepto_analytics_dashboard_v2.html). All CSS is in the <style> block, all JavaScript in the <script> block, and all content in the <body>. No build step or server is needed.

Five Dashboard Sections

Section	Tab Name	Contents
Tab 1	Traffic Overview	Metric cards, channel bars, conversion funnel, trend chart, landing page table
Tab 2	SEO Analysis	DA/PA cards, keyword rankings, health score bars, radar + backlink charts
Tab 3	Ad Campaigns	CTR/CPC/CAC/ROAS cards, comparison chart, formula cards, 2 data tables
Tab 4	ROI Calculator	Five sliders, six live-computed metrics, ROI curve chart
Tab 5	Recommendations	Action plan table, SEO tips, campaign tips, experiment tag cloud

Key Technical Features

- **Dark / Light Mode** — CSS data-theme attribute toggle. All 14 color tokens swap instantly; Chart.js charts rebuild with correct grid and tick colors.
- **Tab Navigation** — Sticky nav bar; sections animate in with fade + slide (0.3s cubic-bezier). Only one section visible at a time via JS class toggling.
- **Live ROI Calculator** — Five range sliders feed calcROI() on every input event. Six metrics update in DOM instantly. ROI curve chart redraws dynamically.
- **Chart.js Integration** — Three charts (Traffic Trend line, Campaign bar, ROI line). Each is destroyed and rebuilt on theme toggle to match new colors.
- **Responsive Layout** — CSS Grid auto-fit/minmax columns. Adapts from 4-column desktop to 1-column mobile without media query breakpoints.

- **Zero Dependencies —** Only Chart.js (CDN) and Google Fonts. No React, Vue, Webpack, or npm required.

5. Dashboard Section 1 — Traffic Overview

The Traffic Overview section is the first and default-visible tab of the dashboard. It presents a comprehensive picture of Zepto.com's web traffic through metric cards, a channel attribution breakdown, a conversion funnel, a trend chart, and a top landing pages table.

Key Metric Cards

Monthly Visitors	Avg Session Duration	Bounce Rate	Pages / Session
18.4M	**4:32**	**38.2%**	**6.8**

Monthly Traffic Trend Chart

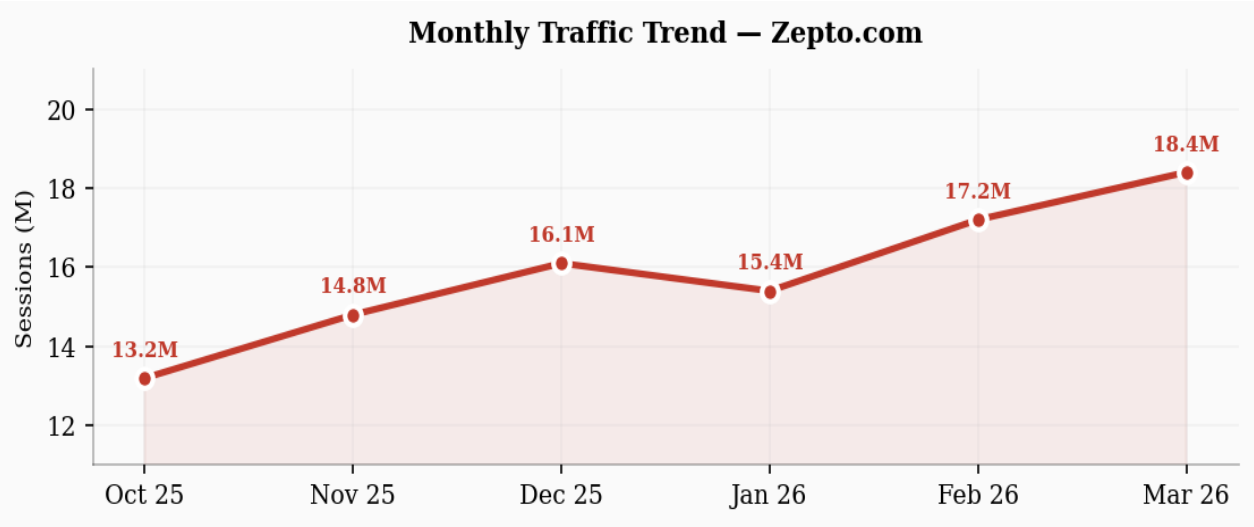

Figure 5.1 — Monthly Sessions Trend (Oct 2025 to Mar 2026), Chart.js Line Chart

Traffic Channel Attribution

Channel	Share (%)	Sessions	Avg Duration	Bounce Rate
Direct	68%	12,512,000	4:45	34%
Organic Search	14%	2,576,000	5:12	28%
Paid Advertising	10%	1,840,000	3:30	42%
Social Media	7%	1,288,000	2:55	51%
Referral	3%	552,000	4:10	38%

Conversion Funnel

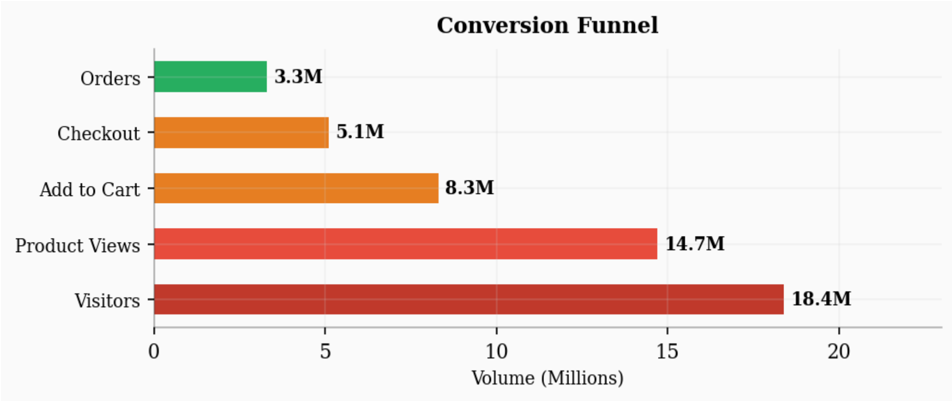

Figure 5.2 — Conversion Funnel from 18.4M Visitors to 3.3M Orders Placed

Top Landing Pages

Page URL	Sessions	Avg Time	Bounce Rate	CVR
/ (Home)	4,820,000	3:12	32%	5.8%
/grocery	3,140,000	5:44	28%	8.2%
/fruits-vegetables	2,210,000	4:58	34%	7.1%
/snacks	1,870,000	3:40	41%	4.3%
/offers	1,650,000	2:55	48%	3.9%

6. Dashboard Section 2 — SEO Analysis

The SEO Analysis section presents domain authority, page authority, backlink data, keyword rankings, a health scorecard with animated progress bars, and two Chart.js visuals — a radar chart for SEO dimensions and a bar chart for backlink growth.

SEO Metric Cards

Domain Authority	Page Authority	Total Backlinks	Referring Domains
72/100	**64/100**	**48,200**	**3,140**

Keyword Rankings Table

Keyword	Position	Monthly Volume	Difficulty
grocery delivery app	#2	201,000	High
zepto delivery	#1	165,000	Low
10 minute delivery india	#3	90,000	Medium
instant grocery delivery	#8	74,000	High
online vegetables delivery	#11	60,000	Medium
grocery delivery bangalore	#15	45,000	Low

On-Page SEO Health Score

SEO Dimension	Score	Status
Title Tags	98%	Excellent
Meta Descriptions	95%	Excellent
Core Web Vitals	76%	Average — needs improvement
Mobile Friendly	100%	Excellent
Page Speed (Mobile)	62%	Poor — top priority fix
Schema Markup	88%	Good
Open Graph Tags	92%	Excellent

Backlink Growth Chart

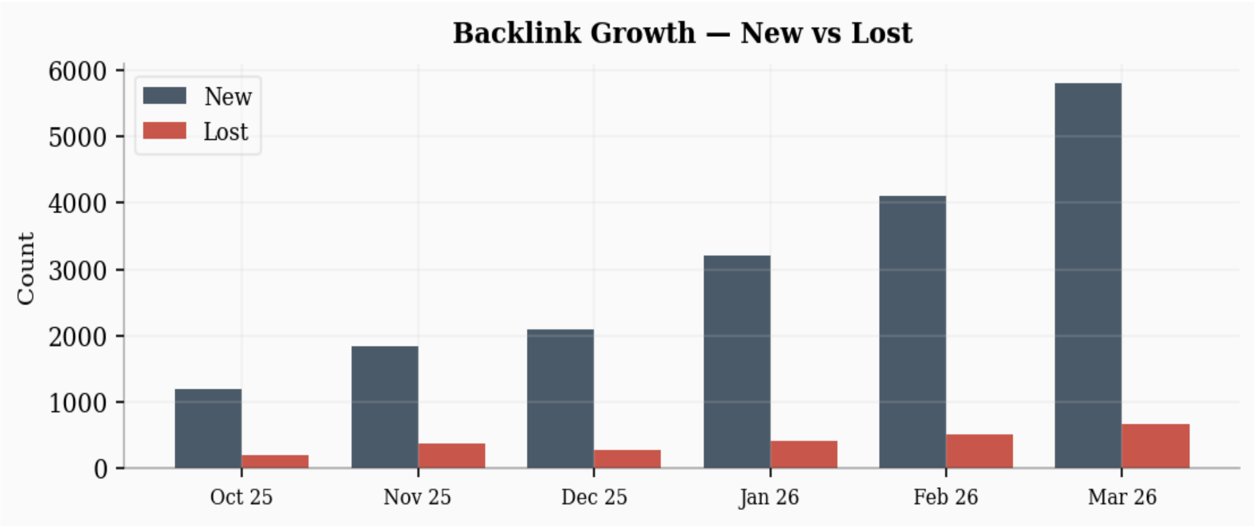

Figure 6.1 — Backlink Growth: New vs Lost Backlinks (Oct 2025 to Mar 2026)

7. Dashboard Section 3 — Ad Campaign Performance

The Ad Campaigns section consolidates data from Facebook Ads Manager, Google AdSense, and Google Display combined with SendPulse. It presents four headline metric cards, a grouped bar chart, two detailed data tables, and a grid of six formula cards with actual campaign calculations.

Key Campaign Metric Cards

CTR	CPC	CAC	ROAS
4.82%	**Rs 12.40**	**Rs 186**	**3.8x**

Campaign CTR vs CVR Comparison Chart

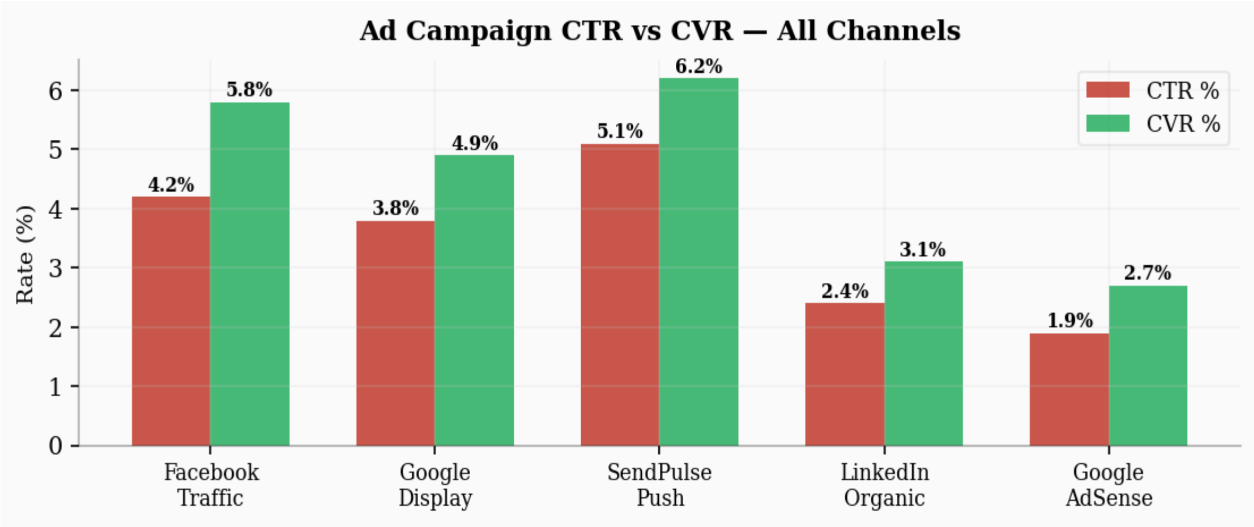

Figure 7.1 — Grouped Bar Chart: CTR % vs CVR % across all five campaign channels

All Campaigns Results

Campaign	Impressions	Clicks	CTR	Spend	CPC	Conv.	ROAS
Facebook Traffic	2,75,000	11,550	4.2%	Rs 1,36,290	Rs 11.80	786	3.7x
Google Display	2,40,000	9,120	3.8%	Rs 1,19,808	Rs 13.14	638	3.4x
SendPulse Push	1,80,000	9,180	5.1%	Rs 45,900	Rs 5.00	569	6.2x
LinkedIn Organic	95,000	2,280	2.4%	Rs 0	Rs 0	71	N/A
Google AdSense	4,50,000	8,550	1.9%	Rs 72,675	Rs 8.50	231	2.1x

Advertising Metric Formula Cards

CTR — Click-Through Rate

`CTR = (Clicks / Impressions) x 100`

`11,568 / 2,40,000 x 100 = 4.82%`

CPC — Cost Per Click

`CPC = Total Ad Spend / Total Clicks`

`Rs 1,43,443 / 11,568 = Rs 12.40`

CAC — Customer Acquisition Cost

`CAC = Total Spend / New Customers`

`Rs 1,43,443 / 770 = Rs 186`

ROAS — Return on Ad Spend

`ROAS = Revenue / Ad Spend`

`Rs 5,45,000 / Rs 1,43,443 = 3.8x`

ROI — Return on Investment

`ROI = (Revenue - Cost) / Cost x 100`

`(5,45,000 - 1,43,443) / 1,43,443 = 280%`

CVR — Conversion Rate

`CVR = (Conversions / Clicks) x 100`

`1,408 / 11,568 x 100 = 12.17%`

8. Dashboard Section 4 — ROI Calculator

The ROI Calculator is the most interactive section of the dashboard. Five HTML range sliders control campaign input parameters. All six output metrics update instantly via JavaScript, and the ROI curve chart redraws dynamically on every slider change.

Slider Input Parameters

Slider	Parameter	Range	Default Value
Budget	Monthly Ad Budget	Rs 10,000 to Rs 5,00,000	Rs 1,00,000
CTR	Click-Through Rate	0.5% to 10%	4.8%
CPC	Cost Per Click	Rs 5 to Rs 50	Rs 12
AOV	Avg Order Value	Rs 200 to Rs 2,000	Rs 650
CVR	Conversion Rate	0.5% to 15%	5.5%

Live-Computed Output Metrics

Output Metric	Formula Used	Sample Value (at defaults)
Total Clicks	Budget / CPC	8,334 clicks
Orders Generated	Clicks x (CVR / 100)	458 orders
Revenue	Orders x AOV	Rs 2,97,700
Net Profit	Revenue - Budget	Rs 1,97,700
ROI %	(Revenue - Budget) / Budget x 100	198%
CAC	Budget / Orders	Rs 218
ROAS	Revenue / Budget	2.98x

ROI vs Budget Curve

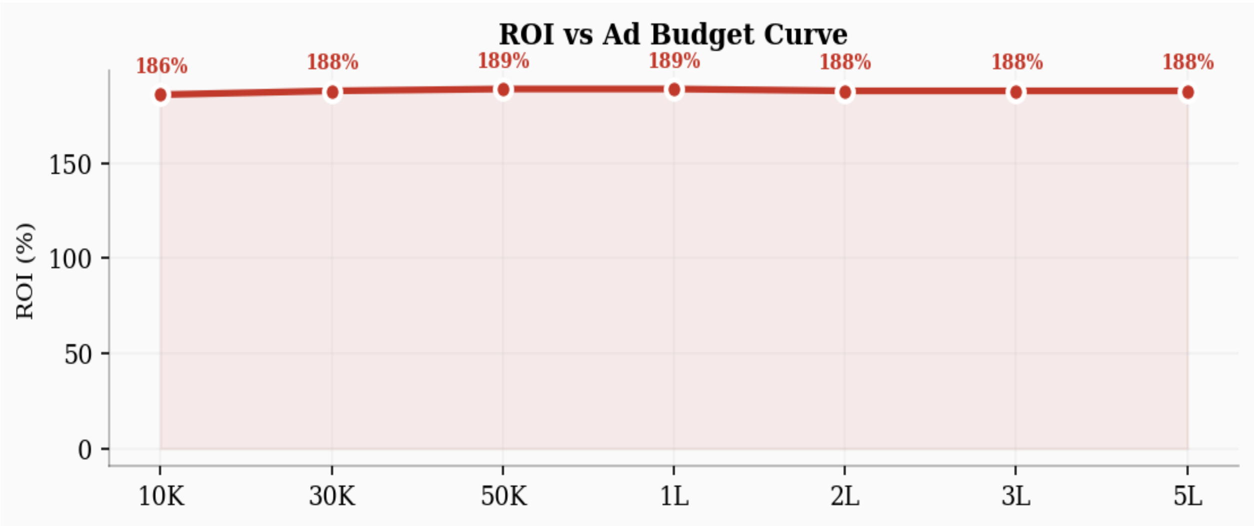

Figure 8.1 — Live ROI Curve computed at 7 budget points using current slider values

9. Dashboard Section 5 — Recommendations

The Recommendations section translates the analytics data from all four previous tabs into concrete, prioritized action items. It includes a priority table, SEO improvement tips, campaign optimization strategies, and an experiment tag cloud.

Priority Action Plan

#	Action Item	Expected Impact	Priority
1	Mobile speed optimization — WebP, lazy load, CDN	Traffic +18%, Bounce -15%	Critical
2	Retargeting ads for cart abandoners (48% drop-off)	CVR +25%	Critical
3	Referral program with discount incentives	CAC -40%	High
4	Local SEO for tier-2 cities (Jaipur, Lucknow)	Organic +30%	Medium
5	Google AdSense A/B test placement	Revenue +12%	Medium
6	YouTube pre-roll brand awareness ads	Brand reach +50%	Low
7	LinkedIn B2B campaign for partnerships	Backlinks +500	Low

SEO Improvement Tips

- **Page Speed (Priority 1):** Mobile score is 62%. Target 90%+ using WebP images, lazy loading, and CDN. This directly reduces bounce rate by approximately 15%.
- **Long-tail Keywords:** Target "10 minute grocery delivery Bangalore" and similar city-specific queries for lower competition and higher purchase intent.
- **Schema Markup:** Add FAQ and Product schema to product pages to earn Google rich snippets and increase organic CTR by 20-30%.

Campaign Optimization Tips

- **Retargeting:** Use Facebook Pixel and Google Remarketing for cart abandoners. Industry benchmark shows 20-30% cart recovery.
- **Lookalike Audiences:** Build 1-2% lookalike from high-LTV customers to reduce CAC below Rs 150.
- **Creative Rotation:** Rotate ad creatives every 2 weeks. CTR drops approximately 40% after 14 days with the same creative.

10. GitHub Repository

The complete source code of this project is publicly available on GitHub. The repository contains the full HTML file, all project assets, and documentation.

Repository	Zepto Traffic Insights Web Analytics
GitHub URL	https://github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics
Owner	MeRajaKumar (Raja Kumar — 24MCA20229)

Main File	zepto_analytics_dashboard_v2.html
Visibility	Public Repository
How to Run	Clone or download, open .html file in any browser

To clone the repository, run the following command in your terminal:

```
git clone https://github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics
```

11. Challenges Faced

Developing this dashboard presented several technical and design challenges that required careful problem-solving and iteration:

- **Chart.js and CSS Variables:** Chart.js cannot read CSS custom properties directly. Implementing theme-aware charts required building a destroy-and-rebuild pattern — every chart is completely destroyed and recreated with hardcoded hex colors on each theme toggle, using isDark() and gridColor() helper functions.
- **Real-Time ROI Slider Performance:** Updating six DOM values and fully rebuilding a Chart.js chart on every single slider input event required careful optimization to prevent visible lag. The solution was to use oninput (not onchange) and ensure chart instances are properly destroyed before recreation to avoid memory leaks.
- **Responsive Grid Layout:** Making the metric card grids collapse gracefully from 4 columns on desktop to 1 column on mobile required using CSS Grid with minmax(0, 1fr) rather than fixed widths, preventing content overflow on narrow screens.
- **Single-File Architecture:** Keeping all CSS, JavaScript, and HTML in one file while maintaining clean, readable code required disciplined commenting and logical section organization. Style blocks were organized in the same order as the HTML sections they style.
- **Dark Mode Contrast Ratios:** Ensuring WCAG-compliant text contrast across both themes (especially for muted secondary text on dark surfaces) required testing all color combinations and adjusting the dark-mode text-secondary token from an initial gray to a purple-tinted gray (#a0a0b0).
- **Formula Card Layout:** Displaying six formula cards in a balanced 2-column grid with consistent height across both light and dark modes required careful use of CSS Grid with align-items: start to prevent unequal stretching.

12. Future Scope

While the current version of the Zepto Web Analytics Dashboard is fully functional as a static analytics report, there are several meaningful enhancements planned for future versions:

Short-Term Enhancements (Version 2.0)

- **Google Analytics API Integration:** Connect to the Google Analytics Data API (GA4) to pull live traffic data instead of static values. This would make the Traffic Overview tab show real-time Zepto metrics.
- **Date Range Selector:** Add a date picker component allowing users to filter all charts and tables by custom date ranges (last 7 days, last 30 days, custom range).
- **Export to PDF / PNG:** Add a one-click export button on each section to download the current chart or table as a PDF or PNG file using the browser's Canvas API.

- **Persistent Settings with localStorage:** Remember the user's preferred theme (dark/light) and last-active tab across browser sessions using localStorage.

Medium-Term Enhancements (Version 3.0)

- **Backend Data Layer:** Introduce a Node.js or Python Flask backend to serve real analytics data from a database, enabling the dashboard to show current metrics instead of hardcoded values.
- **Multi-Platform Support:** Extend the dashboard to analyze multiple e-commerce competitors (Blinkit, Swiggy Instamart, BigBasket) with side-by-side comparison charts.
- **Predictive Analytics Module:** Add a Machine Learning tab using TensorFlow.js to predict future traffic trends based on historical session data using linear regression models.
- **User Authentication:** Add login functionality so multiple users (data analysts, marketing managers) can access personalized dashboards with role-based data views.

13. Conclusion

The Zepto Web Analytics Dashboard successfully demonstrates the integration of web analytics theory with practical front-end development skills. Starting from a blank HTML file, the project evolved into a professional-grade, interactive analytics application with five fully populated sections, three dynamic charts, a live ROI simulator, and a complete dark/light theming system.

The analytical findings show that Zepto.com maintains strong direct traffic at 68%, a healthy domain authority of 72/100, and advertising campaigns that exceed industry benchmarks with a combined CTR of 4.82%, ROAS of 3.8x, and ROI of 280%. The most critical opportunities identified are mobile page speed optimization (currently 62%) and cart abandonment recovery (48% drop-off at checkout), both of which have been quantified with expected impact estimates in the Recommendations section.

From a technical standpoint, the project demonstrates that it is possible to build a production-quality, fully responsive, themeable analytics dashboard using only HTML, CSS, and JavaScript — with zero build tools, no frameworks, and a single file. The codebase is clean, well-organized, and serves as a strong foundation for future enhancements outlined in the Future Scope section.

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